

LECTURE I: CELLULAR INJURY

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CELL INJURY

DEFINATION

NORMAL CELL

ETIOLOGY OF CELL INJURY

PATHOGENESIS OF CELL INJURY

MORPHOLOGY OF CELL INJURY

CELL INJURY

- Cells are the basic units of tissues, which form organs and systems in the human body.
- *Cell injury is defined as a variety of stresses a cell encounters as a result of changes in its internal and external environment.*
- The cellular response to stress may vary and depends upon the following variables:
 - i) The type of cell and tissue involved.
 - ii) Extent and type of cell injury.

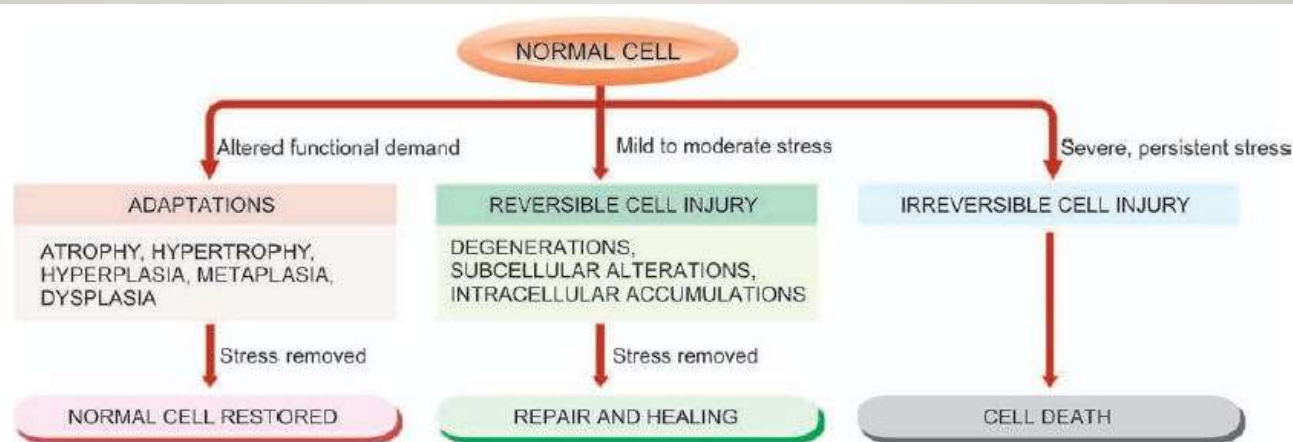


Figure 3.1 ♦ Cellular responses to cell injury.

THE NORMAL CELL

ETIOLOGY OF CELL INJURY

- The cells may be broadly injured by two major ways:
- A. By genetic causes
- B. By acquired causes
- GENETIC CAUSES:-
 - Development defect: error on morphogenesis
 - Cytogenetic defect: chromosomal abnormalities

• ACQUIRED CAUSES:-

1. Hypoxia and ischemia(Deficiency of oxygen)
2. Physical agents
3. Chemical agents and drugs
4. Microbial agents
5. Immunologic agents
6. Nutritional derangements
7. Aging
8. Psychogenic diseases

I. HYPOXIA AND ISCHAEMIA

- Cells of different tissues essentially require oxygen to generate energy and perform metabolic functions.
- Hypoxia is the most common cause of cell injury. Hypoxia may result from the following:
 1. Cell injury is by reduced supply of blood to cells due to interruption.
 2. Disorders of oxygen-carrying RBCs (e.g. anaemia, carbon monoxide poisoning), heart diseases, lung diseases and increased demand of tissues.

2. PHYSICAL AGENTS

Physical agents in causation of disease are as under:

1. Mechanical trauma (e.g. road accidents)
2. Thermal trauma (e.g. by heat and cold)
3. Electricity
4. Radiation (e.g. ultraviolet and ionising)
5. Rapid changes in atmospheric pressure.

3. **CHEMICALS AND DRUGS**

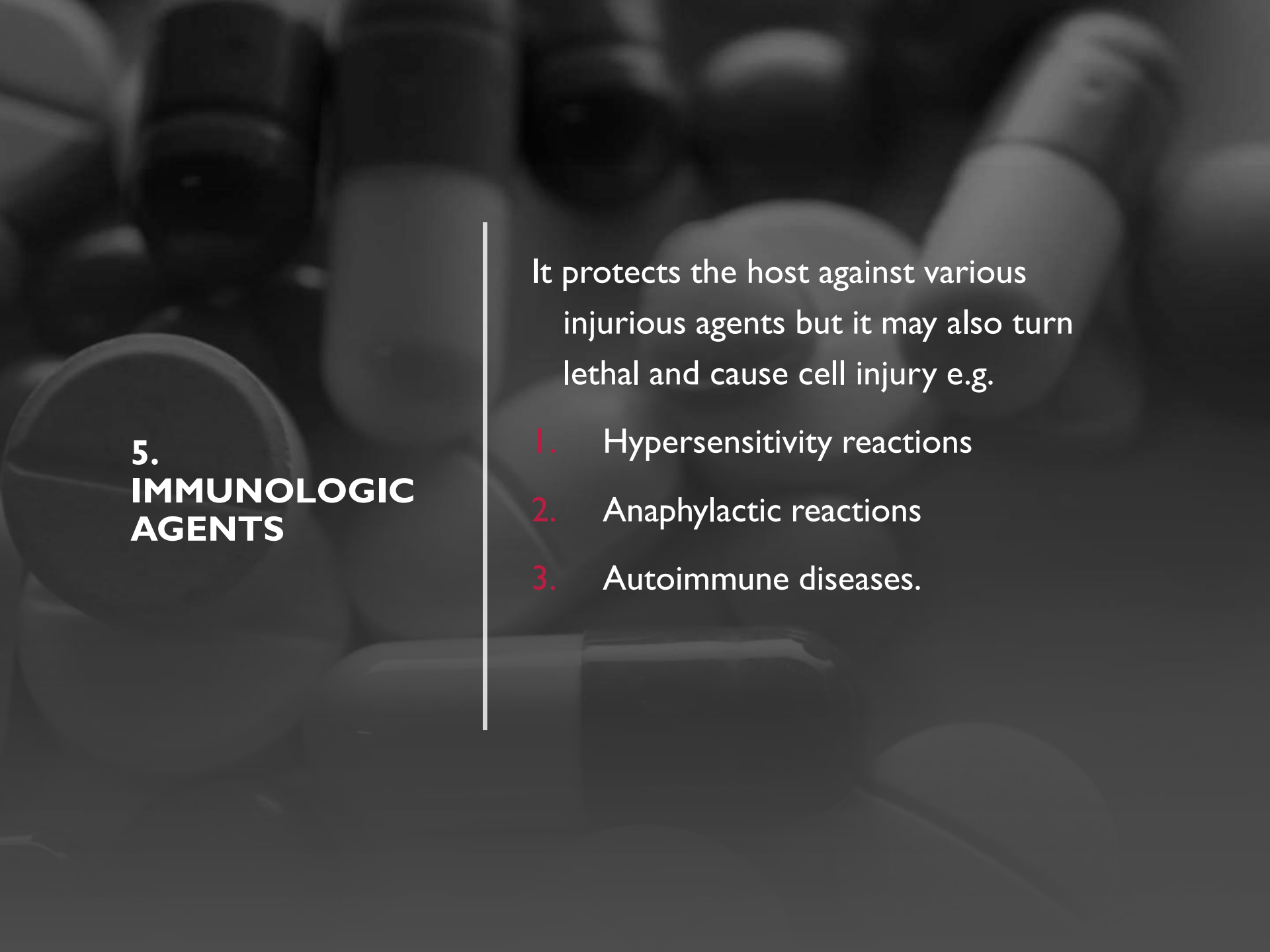
An ever increasing list of chemical agents and drugs may cause cell injury

1. chemical poisons such as cyanide, arsenic, mercury
2. strong acids and alkalis
3. environmental pollutants
4. insecticides and pesticides
5. oxygen at high concentrations
6. hypertonic glucose and salt
7. social agents such as alcohol and narcotic drugs
8. therapeutic administration of drugs

4. **MICROBIAL AGENTS**

Injuries by microbes include infections
caused by

1. Bacteria,
2. Rickettsiae
3. Viruses
4. Fungi
5. Protozoa
6. Metazoa
7. Parasites



5. IMMUNOLOGIC AGENTS

It protects the host against various injurious agents but it may also turn lethal and cause cell injury e.g.

1. Hypersensitivity reactions
2. Anaphylactic reactions
3. Autoimmune diseases.

6. NUTRITIONAL DERANGEMENTS

A deficiency of nutrients may result in nutritional imbalances.

1. Deficiency of nutrients (e.g. Starvation)
2. Protein calorie (e.g. Marasmus, Kwashiorkor)
3. Minerals (e.g. Anaemia)
4. Nutritional excess is a problem of affluent societies resulting in obesity, atherosclerosis, heart disease and hypertension.

7. AGING

Cellular aging leads to impaired ability of the cells to undergo replication and repair, and ultimately lead to cell death culminating in death of the individual.

8. PSYCHOGENIC DISEASES

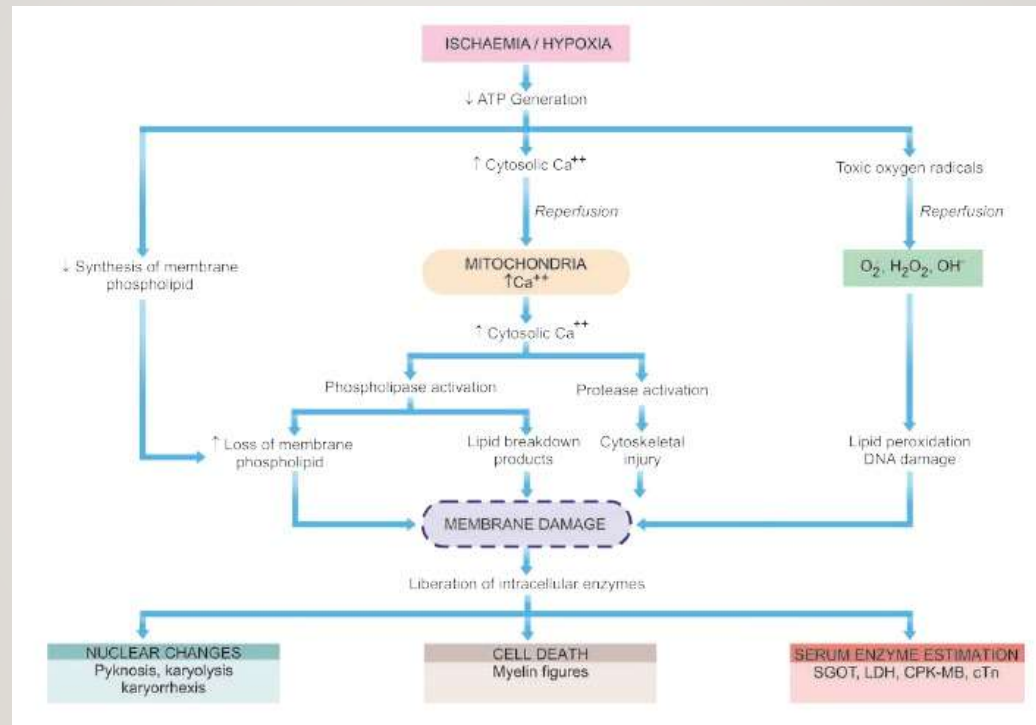
Diseases due to mental stress, strain, anxiety, overwork and frustration e.g. depression, schizophrenia.

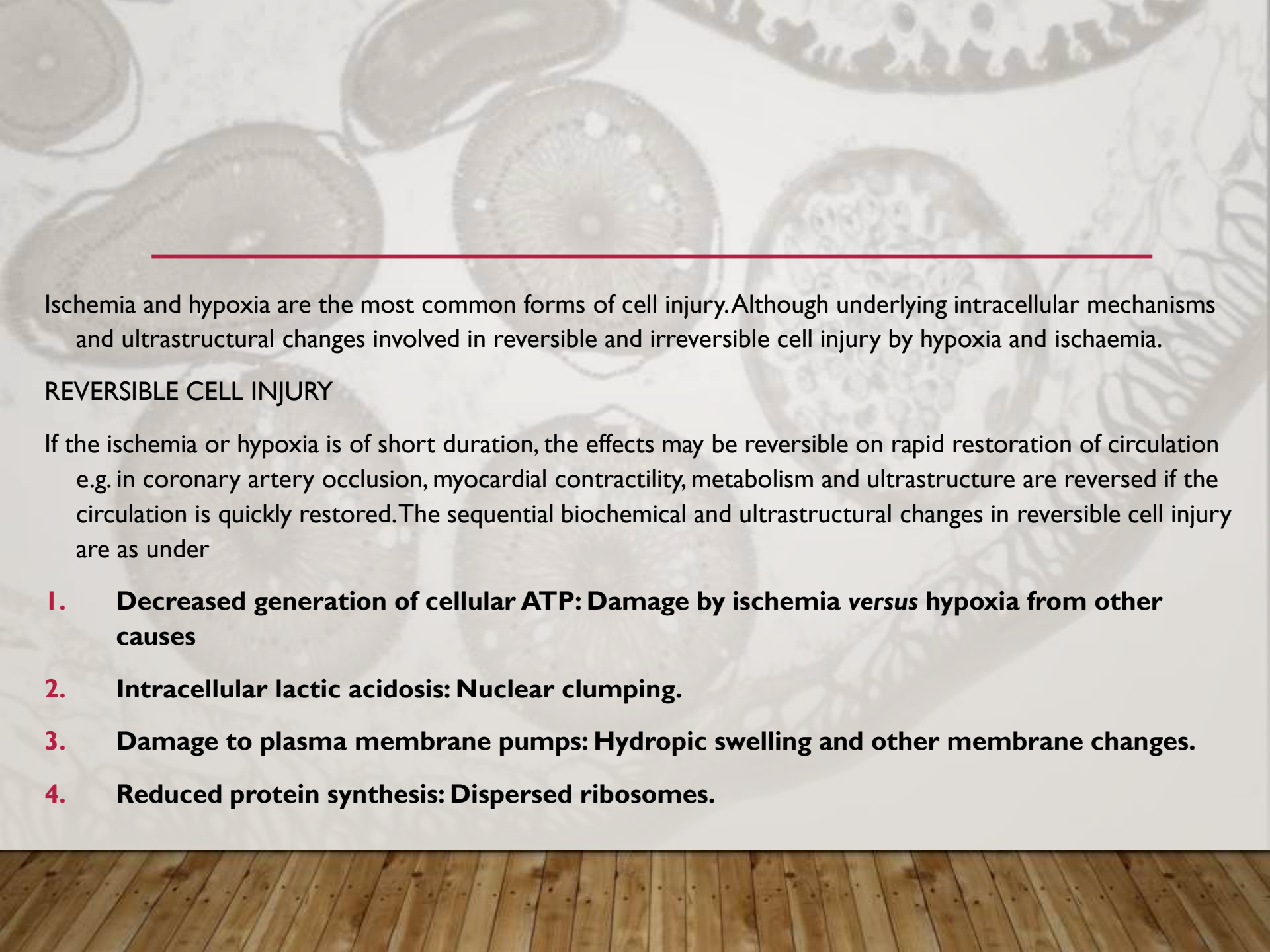


PATHOGENESIS OF CELL INJURY

- The underlying alterations in biochemical systems of cells for reversible and irreversible cell injury by various agents is complex and varied.
- The following *principles apply in pathogenesis* of most forms of cell injury by various agents:
 - **Type, duration and severity of injurious agent**
 - **Type, status and adaptability of target cell**
 - **Underlying intracellular phenomena**
 - **Morphologic consequences**

PATHOGENESIS OF ISCHAEMIC AND HYPOXIC INJURY





Ischemia and hypoxia are the most common forms of cell injury. Although underlying intracellular mechanisms and ultrastructural changes involved in reversible and irreversible cell injury by hypoxia and ischaemia.

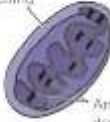
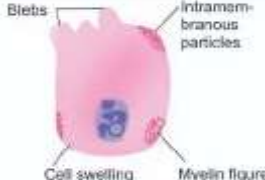
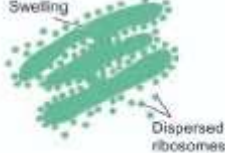
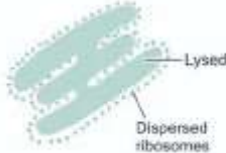
REVERSIBLE CELL INJURY

If the ischemia or hypoxia is of short duration, the effects may be reversible on rapid restoration of circulation e.g. in coronary artery occlusion, myocardial contractility, metabolism and ultrastructure are reversed if the circulation is quickly restored. The sequential biochemical and ultrastructural changes in reversible cell injury are as under

1. **Decreased generation of cellular ATP: Damage by ischemia versus hypoxia from other causes**
2. **Intracellular lactic acidosis: Nuclear clumping.**
3. **Damage to plasma membrane pumps: Hydropic swelling and other membrane changes.**
4. **Reduced protein synthesis: Dispersed ribosomes.**

- **IRREVERSIBLE CELL INJURY**

- Persistence of ischemia or hypoxia results in irreversible damage to the structure and function of the cell (cell death). The stage at which this *point of no return or irreversibility* is reached from reversible cell injury is unclear but the sequence of events is a continuation of reversibly injured cell.
- Two essential phenomena always distinguish irreversible from reversible cell injury
 1. Inability of the cell to reverse *mitochondrial dysfunction* on reperfusion or reoxygenation.
 2. *Disturbance in cell membrane function* in general, and in plasma membrane in particular.
- These biochemical changes have effects on the ultrastructural components of the cell
 1. **Calcium influx: Mitochondrial damage**
 2. **Activated phospholipases: Membrane damage**
 3. **Intracellular proteases: Cytoskeletal damage**
 4. **Activated endonucleases: Nuclear damage**
 5. **Lysosomal hydrolytic enzymes: Lysosomal damage, cell death and phagocytosis.**

ORGANELLES IN NORMAL CELL	A, REVERSIBLE CELL INJURY	B, IRREVERSIBLE CELL INJURY
 1. MITOCHONDRION	 Swelling Amorphous densities	 Swollen with vacuoles Large densities
 2. MEMBRANES	 Blebs Intramembranous particles Cell swelling Myelin figure	 Disruption Myelin figure
 3. RER AND RIBOSOMES	 Swelling Dispersed ribosomes	 Lysed Dispersed ribosomes
 4. LYSOSOMES	 Autophagy	 Swollen, ruptured
 5. CYTOSKELETON	 Aggregated	 Disrupted
 6. NUCLEUS	 Clumped chromatin	 Pyknosis Karyolysis Karyorrhexis

MORPHOLOGY OF CELL INJURY

- The molecular and biochemical mechanisms of various forms of cell injury, we now turn to light microscopic morphologic changes of reversible and irreversible cell injury.

Reversible cell injury	Irreversible cell injury
HYDROPIIC CHANGES	AUTOLYSIS
HYALINE CHANGES	NECROSIS
MUCOID CHANGES	APOPTOSIS
FATTY CHANGES	GANGRENE

MORPHOLOGY OF REVERSIBLE CELL INJURY

- The term degeneration has been used to denote morphology of reversible cell injury.
- Following morphologic forms of reversible cell injury are included under this heading:
 1. Hydropic change
 2. Fatty change
 3. Hyaline change
 4. Muroid change

HYDROPIK CHANGES

- It means accumulation of water within the cytoplasm of the cell.
- **ETIOLOGY:-** The common causes include acute and sub acute cell injury from various etiologic agents such as bacterial toxins, chemicals, poisons, burns, high fever, saline etc.
- **PATHOGENESIS:-** Cloudy swelling results from impaired regulation of sodium and potassium at the level of cell membrane. This results in intracellular accumulation of sodium and escape of potassium. This, in turn, leads to rapid flow of water into the cell to maintain iso-osmotic conditions and hence cellular swelling occurs. Hydropic swelling is an entirely reversible change upon removal of the injurious agent.

HYALINE CHANGES

- Hyaline is a descriptive histologic term for glassy, homogeneous, eosinophilic appearance of material in haematoxylin and eosin-stained sections.
- It may be intracellular or extracellular

INTRACELLULAR HYALINE:- It is mainly seen in epithelial cells. A few examples are as follows:

1. *Hyaline droplets in the proximal tubular epithelial cells in cases of excessive reabsorption of plasma proteins.*
2. *Hyaline degeneration of rectus abdominalis muscle called Zenker's degeneration, occurring in typhoid fever..*
3. *Mallory's hyaline represents aggregates of intermediate filaments in the hepatocytes in alcoholic liver cell injury.*
4. *Nuclear hyaline inclusions seen in some viral infections.*
5. *Russell's bodies representing excessive immunoglobulins in the rough endoplasmic reticulum of the plasma cells*

EXTRACELLULAR HYALINE:- It is seen in connective tissues. A few examples of extracellular hyaline change are as under:

1. Hyaline degeneration in *leiomyomas of the uterus*
2. Hyalinised *old scar of fibrocollagenous tissues*.
3. *Hyaline arteriosclerosis in renal vessels in hypertension and diabetes mellitus.*
4. Hyalinised glomeruli in *chronic glomerulonephritis*.
5. *Corpora amylacea* are rounded masses of concentric hyaline laminae seen in the prostate in the elderly, in the brain and in the spinal cord in old age, and in old infarcts of the lung.

MUCOID CHANGES

- Mucus secreted by mucous glands is a combination of proteins complexed with mucopolysaccharides
- *Mucin*, a glycoprotein,
- Mucin is normally produced by epithelial cells of mucous membranes and mucous glands, as well as by some connective tissues like in the umbilical cord.
- Both types of mucin are stained by alcian blue.
- Epithelial mucin stains positively with periodic acid-Schiff (PAS), while connective tissue mucin is PAS negative but is stained positively with colloidal iron.

EPITHELIAL MUCIN:- following are some examples of functional secretions of epithelial mucin:-

- 1. Catarrhal inflammation of mucous membrane (e.g. of respiratory tract, alimentary tract, uterus).
- 2. Obstruction of duct leading to mucocele in the oral cavity and gallbladder.
- 3. Cystic fibrosis of the pancreas.
- 4. Mucin-secreting tumour.

- **CONNECTIVE TISSUE MUCIN:-** A few examples of disturbances of connective tissue mucin are as under:
 - 1. Muroid degeneration in some tumours e.g. myxomas, neurofibromas, fibroadenoma, soft tissue sarcomas etc
 - 2. Dissecting aneurysm of the aorta due to Erdheim's medial degeneration and Marfan's syndrome.
 - 3. Myxomatous change in the dermis in myxoedema.
 - 4. Myxoid change in the synovium in ganglion on the wrist.

FATTY CHANGES

- It is the intracellular accumulation of neutral fat within parenchymal cells.
- It includes the older, now abandoned, terms of *fatty degeneration* and *fatty infiltration* because *fatty change* neither necessarily involves degeneration nor infiltration.
- It is especially common in the liver but may occur in other non-fatty tissues like the heart, skeletal muscle, kidneys and other organ.

FATTY LIVER

Liver is the commonest site for accumulation of fat because it plays central role in fat metabolism.

ETIOLOGY:- Fatty change in the liver may result from one of the two types of causes:

1. Conditions with excess fat: *exceeding the capacity of the liver to metabolise it.*

i) Obesity

ii) Diabetes mellitus

iii) Congenital hyperlipidaemia

2. Liver cell damage: *when fat cannot be metabolised in it*

i) Alcoholic liver disease

ii) Starvation

iii) Protein calorie malnutrition

iv) Chronic illnesses (e.g. tuberculosis)

v) Acute fatty liver in late pregnancy

vi) Hypoxia (e.g. anaemia, cardiac failure)

vii) Hepatotoxins (e.g. carbon tetrachloride etc.)

viii) Drug-induced liver cell injury (e.g. methotrexate, steroids, CCl₄, etc)

ix) Reye's syndrome

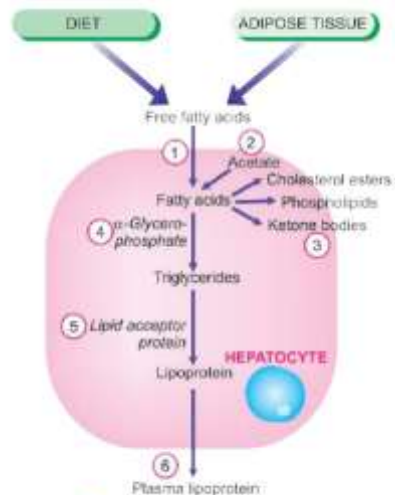


Figure 3.16 Lipid metabolism in the pathogenesis of fatty liver.

- **PATHOGENESIS:-** Mechanism of fatty liver depends upon the stage at which the etiologic agent acts in the normal fat transport and metabolism. Hence, pathogenesis of fatty liver is best understood in the light of normal fat metabolism in the liver .
- Lipids as free acids enter the liver cell from either of the following 2 sources
 1. From diet as chylomicrons
 2. From adipose tissue as free fatty acids.

- **In fatty liver, intracellular accumulation of triglycerides** can occur due to defect at one or more of the following 6 steps *in the normal fat metabolism shown in*

1. Increased entry of free fatty acids into the liver.
2. Increased synthesis of fatty acids by the liver.
3. Decreased conversion of fatty acids into ketone bodies resulting in increased esterification of fatty acids to triglycerides.
4. Increased α -glycerophosphate causing increased esterification of fatty acids to triglycerides.
5. Decreased synthesis of 'lipid acceptor protein' resulting in decreased formation of lipoprotein from triglycerides.
6. Block in the excretion of lipoprotein from the liver into plasma.

MORPHOLOGY OF IRREVERSIBLE CELL INJURY (CELL DEATH)

- Cell death is a state of irreversible injury.
 - It may occur in the living body as a local change (i.e. autolysis, necrosis and apoptosis) and the changes that follow it (i.e. gangrene and pathologic calcification), or result in end of the life.
 - processes involved in cell death are described below.
1. AUTOLYSIS
 2. NECROSIS
 3. APOPTOSIS
 4. GANGRENE

AUTOLYSIS

- Autolysis (i.e. *self-digestion*) is *disintegration of the cell by its own hydrolytic enzymes liberated from lysosomes*.
- It can occur in the living body when it is surrounded by inflammatory reaction *but the term is generally used for postmortem change in which there is complete absence of surrounding inflammatory response*.
- It is *rapid in some tissues rich in hydrolytic enzymes such as in the pancreas, and gastric mucosa; intermediate in tissues like the heart, liver and kidney; and slow in fibrous tissue*.
- Morphologically, it is identified by homogeneous and eosinophilic cytoplasm with loss of cellular details and remains of cell as debris.

NECROSIS

- It is defined as localised area of death of tissue followed by degradation of tissue by hydrolytic enzymes liberated from dead cells; it is invariably accompanied by inflammatory reaction.
- It can be caused by various agents such as hypoxia, chemical and physical agents, microbial agents, immunological injury, etc.
- Two essential changes characterise irreversible cell injury in necrosis of all types
 1. Cell digestion by lytic enzymes.
 2. Denaturation of proteins.

TYPES OF NECROSIS

1. **COAGULATIVE NECROSIS**
2. **LIQUEFACTION NECROSIS**
3. **CASEOUS NECROSIS**
4. **FAT NECROSIS**
5. **FIBRINOID NECROSIS**

1. COAGULATIVE NECROSIS:- It caused by irreversible focal injury, mostly from sudden cessation of blood flow and less often from bacterial and chemical agents. The organs commonly affected are the heart, kidney, and spleen.

2. LIQUEFACTION NECROSIS:- It occurs commonly due to ischaemic injury and bacterial or fungal infections. It occurs due to degradation of tissue by the action of powerful hydrolytic enzymes. The common examples are infarct brain and abscess cavity.

3. CASEOUS NECROSIS:- It is found in the centre of foci of TB infections. It combines features of both coagulative and liquefactive necrosis.

4. FAT NECROSIS:- It is a special form of cell death occurring at two anatomically different locations but morphologically similar lesions. These are

- *Acute pancreatic necrosis*
- *traumatic fat necrosis*
- In the case of pancreas, there is liberation of pancreatic lipases from injured or inflamed tissue that results in necrosis of the pancreas as well as of the fat depots throughout the peritoneal cavity.
- Fat necrosis hydrolyses neutral fat present in adipose cells into glycerol and free fatty acids. The damaged adipose cells assume cloudy appearance. The leaked out free fatty acids complex with calcium to form calcium soaps.

5. FIBRINOID NECROSIS:- It is characterized by deposition of fibrin-like material which has the staining properties of fibrin. It is encountered in examples of immunologic tissue injury, arterioles in hypertension, peptic ulcer etc.

APOPTOSIS

- It is a form of 'coordinated and internally programmed cell death' having significance in a variety of physiologic and pathologic conditions (*apoptosis is a Greek word meaning 'falling off' or 'dropping off'*).
- The term was first introduced in 1972 as distinct from necrosis by being a form of cell death which is controlled and regulated by the rate of cell division; when the cell is not needed, pathway of cell death is activated ('cell suicide') and is unaccompanied by any inflammation and collateral tissue damage.

APOPTOSIS IN BIOLOGIC PROCESSES:-

Apoptosis is responsible for mediating cell death

in a wide variety of physiologic and pathologic processes as under:

- **Physiologic Processes:**

1. Organized cell destruction in sculpting of tissues during *development of embryo*.
2. Physiologic involution of cells in *hormone-dependent tissues*
3. Normal cell destruction followed by *replacement proliferation such as in intestinal epithelium*.
4. *Involution of the thymus in early age*.

- **Pathologic Processes:**

1. Cell death in tumours exposed to *chemotherapeutic agents*.
2. *Cell death by cytotoxic T cells in immune mechanisms*
3. *Progressive depletion of CD4+T cells in the pathogenesis of AIDS.*
4. *Cell death in viral infections e.g. viral hepatitis.*
5. *Pathologic atrophy of organs and tissues on withdrawal of stimuli e.g. prostatic atrophy, atrophy of kidney*
6. *Cell death in response to injurious agents involved in causation of necrosis e.g. radiation, hypoxia and mild thermal injury.*
7. *In degenerative diseases of CNS e.g. in Alzheimer's disease, Parkinson's disease, and chronic infective dementias.*
8. *Heart diseases e.g. heart failure, acute myocardial infarction*



INITIATORS OF APOPTOSIS

- Transmembranous (e.g. absence of hormones, growth factors, cytokines)
- Intracellular (e.g. heat, radiation, hypoxia)
- Extracellular (e.g. FAS receptor activation)



PROCESS OF
PROGRAMMED CELL DEATH

- Activation of caspases
- Activation of death receptors (*CD95*)
- Activation of growth controlling genes (*BCL-2*, *p53*)
- Cell death



PHAGOCYTOSIS

- Membrane changes (e.g. expression of phosphatidylserine, thrombospondin)
- No inflammatory cells

Figure 3.31  Molecular mechanism of apoptosis.

MOLECULAR MECHANISMS OF APOPTOSIS Several physiologic and pathologic processes activate apoptosis in a variety of ways. However, in general the following events sum up the sequence involved in apoptosis:

Initiators of apoptosis:- Triggers for signalling programmed cell death act at the cell membrane, either intracellularly or extracellularly

i) Withdrawal of signals required for normal cell survival

(e.g. absence of certain hormones, growth factors, cytokines).

ii) Extracellular signals triggering of programmed cell death (e.g. activation of FAS receptor belonging to TNF-R family).

iii) Intracellular stimuli e.g. heat, radiation, hypoxia etc.

Process of programmed cell death:- After the cell has been initiated into self-destruct mode, the programme inbuilt in the cell gets activated as under:

i) *Activation of caspases*

ii) *Activation of death receptors*

iii) *Activation of growth controlling genes (BCL-2 and p53)*

iv) *Cell death.*

Phagocytosis:- The dead apoptotic cells develop membrane changes which promote their phagocytosis. Phosphatidylserine and thrombospondin molecules which are normally present on the inside of the cell membrane, appear on the outer surface of the cells in apoptosis, which facilitate their identification by adjacent phagocytes and promotes phagocytosis.

GANGRENE

- It is a form of necrosis of tissue with superadded putrefaction.
- Coagulative necrosis due to ischaemia (e.g. in gangrene of the bowel, gangrene of limb).
- *Gangrenous inflammation* is characterised by primarily inflammation provoked by virulent bacteria resulting in massive tissue necrosis.
- The end-result of necrotising inflammation and gangrene is the same but the way the two are produced.
- Examples of necrotising inflammation are: gangrenous appendicitis, gangrenous stomatitis.
- There are 2 main forms of gangrene—
 1. dry gangrene
 2. wet gangrene
 3. variant form of wet gangrene called gas gangrene.

DRY GANGRENE

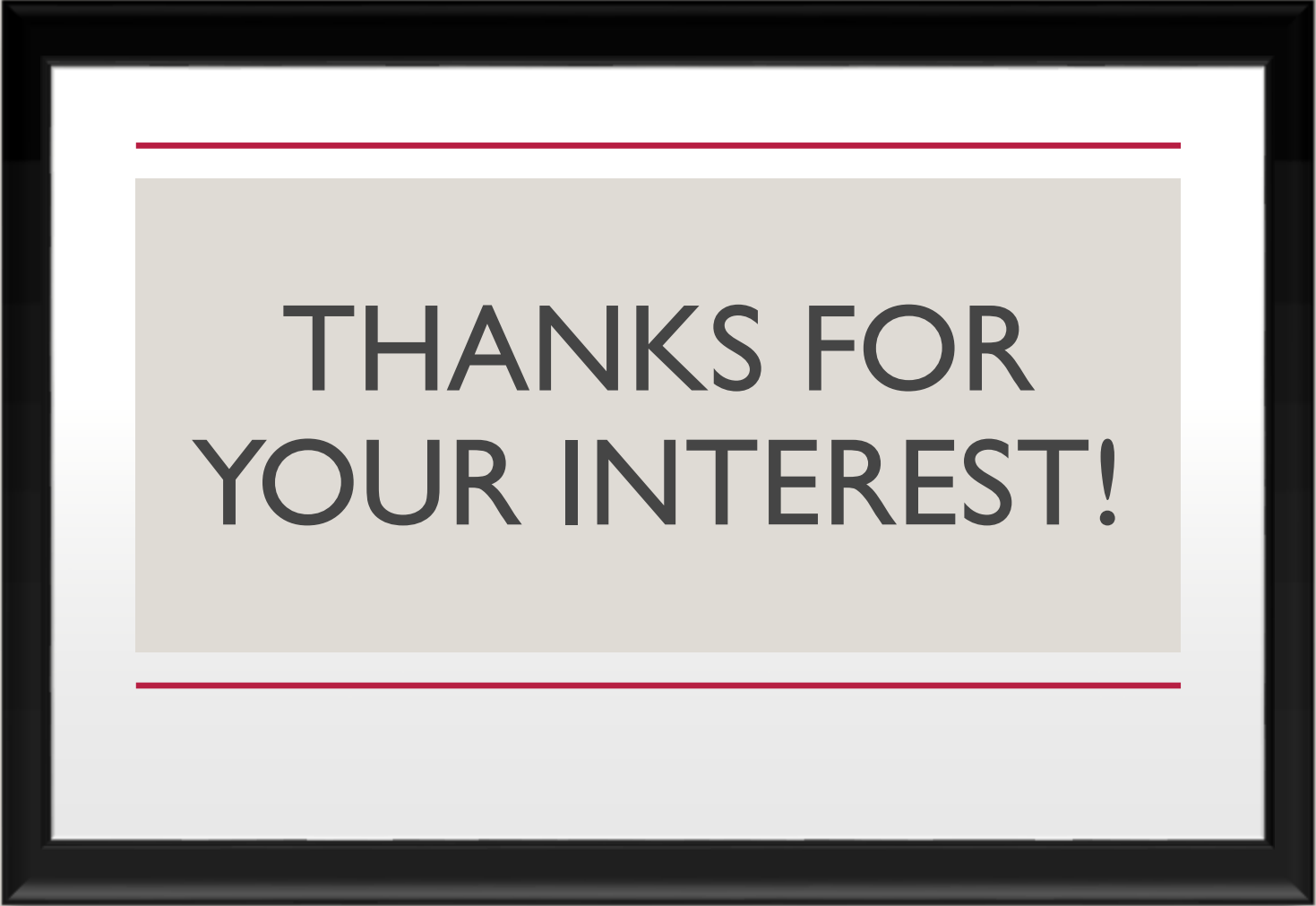
- It begins in the distal part of a limb due to ischaemia.
- Example is the dry gangrene in the toes and feet of an old patient due to arteriosclerosis.
- Other causes of dry gangrene foot include
 1. Buerger's disease
 2. Raynaud's disease
 3. trauma
 4. ergot poisoning
- Gangrene spreads slowly upwards until it reaches a point where the blood supply is adequate to keep the tissue viable.
- *A line of separation is formed at this point between the gangrenous part and the viable part.*

WET GANGRENE

- It occurs in naturally moist tissues and organs such as the mouth, bowel, lung, cervix, vulva etc.
- Diabetic foot is another example of wet gangrene due to high sugar content in the necrosed tissue which favours growth of bacteria.
- *Bed sores occurring in a bed-ridden patient due to pressure on sites like the sacrum, buttocks and heels are the other important clinical conditions included in wet gangrene.*
- Wet gangrene usually develops rapidly due to blockage of venous, and less commonly, arterial blood flow from thrombosis or embolism.
- The affected part is stuffed with blood which favours the rapid growth of putrefactive bacteria. The toxic products formed by bacteria are absorbed causing profound systemic manifestations of septicaemia, and finally death.
- The spreading wet gangrene generally lacks clear-cut line of demarcation and may spread to peritoneal cavity causing peritonitis.

GAS GANGRENE

- It is a special form of wet gangrene caused by gas-forming clostridia (gram-positive anaerobic bacteria) which gain entry into the tissues through open contaminated wounds, especially in the muscles, or as a complication of operation on colon which normally contains clostridia.
- Clostridia produce various toxins which produce necrosis and oedema locally and are also absorbed producing profound systemic manifestations.



THANKS FOR
YOUR INTEREST!